Triangles with Mixed Rooted Branches: A Degree-Biased Jensen Bound

Abstract

For integers $r, s \geq 0$, let $R_{r,s}$ be a triangle with a distinguished vertex o, with r pendant edges and s internally disjoint two-edge paths attached at o. We prove that every graphon of edge density $p \geq 1/2$ satisfies

$$t(R_{r,s}, W) \geq p^{r+2s+1}(2p-1) = \Phi_{R_{r,s}}(p).$$

The cases $s = 0$ and $r = 0$ were known separately. The new point is that the two branch types may occur simultaneously. A complement-kernel symmetrization proves that the mean two-edge-tail density under the measure biased by d^r is at least p^2. A single convex Jensen step then treats all s tails at once. In the six-vertex catalogue this proves the formerly open Atlas graph 95.

1 Definitions and the target

Let (Ω, μ) be an arbitrary probability space, and let $W\colon \Omega^2 \to [0,1]$ be a symmetric measurable graphon. Write

$$p = \int_{\Omega^2} W(x,y)\,\mathrm{d}\mu(x)\,\mathrm{d}\mu(y), \qquad d(x) = \int_{\Omega} W(x,y)\,\mathrm{d}\mu(y),$$

and put

$$A(x) = (T_W d)(x) = \int_{\Omega} W(x,y) d(y)\,\mathrm{d}\mu(y).$$

Thus $A(x)$ is the rooted density of a two-edge path whose initial vertex is fixed at x. Define the rooted triangle density

$$\tau(x) = \int_{\Omega^2} W(x,y)W(x,z)W(y,z)\,\mathrm{d}\mu(y)\,\mathrm{d}\mu(z).$$

For $r, s \geq 0$, construct $R_{r,s}$ from a triangle rooted at o by adjoining r distinct leaves adjacent to o, and s pairwise vertex-disjoint paths $o - u_i - v_i$. No other vertices are identified and no other edges are added. Integrating the branch vertices first gives the exact identity

$$t(R_{r,s}, W) = \int_{\Omega} \tau(x) d(x)^r A(x)^s\,\mathrm{d}\mu(x). \tag{1}$$

The graph has $3 + r + 2s$ vertices and the same number of edges. Once the triangle is properly coloured, every branch vertex has $x - 1$ choices after its parent is coloured. Hence

$$\chi_{R_{r,s}}(x) = x(x-1)^{r+2s+1}(x-2), \qquad \chi(R_{r,s}) = 3.$$

For $q = 1 - p > 0$, direct substitution gives

$$\begin{aligned}\Phi_{R_{r,s}}(p) &= q^{3+r+2s}\frac{1}{q}\left(\frac{p}{q}\right)^{r+2s+1}\left(\frac{2p-1}{q}\right)\\ &= p^{r+2s+1}(2p-1).\end{aligned} \tag{2}$$

This polynomial also gives the continued value 1 at $p = 1$.

2 The weighted tail mean

Put $M_j = \int_\Omega d(x)^j \,\mathrm{d}\mu(x)$ for $j \geq 0$. The next lemma is the ingredient that permits the two branch types to be mixed.

Lemma 2.1 (Degree-biased two-edge-tail mean)**.** *For every integer $r \geq 0$ and every graphon of edge density $p \in [1/2, 1]$,*

$$\int_\Omega d(x)^r A(x)\,\mathrm{d}\mu(x) \geq p^2 M_r.$$

Equivalently, under the probability measure

$$\mathrm{d}\nu_r(x) = \frac{d(x)^r}{M_r}\,\mathrm{d}\mu(x),$$

one has $\mathbb{E}_{\nu_r} A \geq p^2$.

Proof. The denominator M_r is positive because $M_r \geq p^r > 0$ for $r \geq 1$ by Jensen, while $M_0 = 1$. Let $U = 1 - W$, let T_U be its integral operator, and put $u = T_U\mathbf{1} = 1 - d$ and $q = 1 - p$. Since

$$T_U d = T_U(\mathbf{1} - u) = u - T_U u,$$

we have

$$A = T_W d = p\mathbf{1} - T_U d = d - q\mathbf{1} + T_U u.$$

Consequently, if $J_r = \int d^r A$, then

$$J_r = M_{r+1} - qM_r + \langle d^r, T_U u\rangle\,. \tag{3}$$

Symmetry of U gives

$$2\left(\langle d^r, T_U u\rangle - \int d^r u^2\right) = \int_{\Omega^2} U(x,y)\big(u(y) - u(x)\big)\big(d(x)^r - d(y)^r\big)\,\mathrm{d}\mu^2.$$

The integrand is nonnegative: $d = 1 - u$, so u and d^r are oppositely ordered. Thus (3) implies

$$J_r \geq M_{r+2} - M_{r+1} + pM_r. \tag{4}$$

Set $m = M_{r+1}/M_r$. Cauchy–Schwarz gives $M_{r+2}/M_r \geq m^2$. Also $m \geq p$: for $r = 0$ this is equality, while for $r \geq 1$ it follows from

$$M_{r+1} - pM_r = \frac{1}{2}\int_{\Omega^2} (d(x)^r - d(y)^r)(d(x) - d(y))\,\mathrm{d}\mu(x)\,\mathrm{d}\mu(y) \geq 0.$$

Since $p \geq 1/2$ in the application, $m \geq p \geq 1 - p$, and therefore

$$\frac{J_r}{M_r} - p^2 \geq m^2 - m + p - p^2 = (m - p)(m - (1 - p)) \geq 0.$$

This proves the assertion. □

3 Rooted triangle inputs

We record the two elementary estimates needed for the main proof.

Lemma 3.1 (Pointwise rooted triangle)**.** *For almost every x,*

$$\tau(x) \geq \max\{2A(x) - p, 0\}.$$

Proof. The elementary inequality $ab \geq a + b - 1$ for $a, b \in [0, 1]$, applied to $W(x, y)$ and $W(x, z)$ and multiplied by $W(y, z) \geq 0$, gives after integration

$$\tau(x) \geq 2 \int_\Omega W(x, y)d(y)\, \mathrm{d}\mu(y) - p = 2A(x) - p.$$

Also $\tau(x) \geq 0$. □

Lemma 3.2 (Weighted rooted triangle)**.** *For every integer $r \geq 0$ and every graphon of edge density $p > 0$,*

$$\int_\Omega d(x)^r \tau(x)\, \mathrm{d}\mu(x) \geq \frac{2p - 1}{p} M_{r+2}.$$

Proof. Put $I_r = \int d^r \tau$ and $J_r = \int d^r A$. Applying $ab \geq a + b - 1$ to the two edges incident with the fixed root, multiplying by $d(x)^r W(y, z)$, and integrating yields

$$I_r \geq 2J_r - pM_r.$$

Using (4),

$$I_r \geq 2M_{r+2} - 2M_{r+1} + pM_r.$$

Subtracting $(2p - 1)M_{r+2}/p$ from the right-hand side leaves

$$\frac{1}{p}\left(M_{r+2} - 2pM_{r+1} + p^2 M_r\right) = \frac{1}{p} \int_\Omega d(x)^r (d(x) - p)^2\, \mathrm{d}\mu(x) \geq 0.$$

□

Lemma 3.3 (Tail convexity)**.** *For $p > 0$ and every integer $s \geq 1$, the function*

$$f_{p,s}(a) = a^s \max\{2a - p, 0\}, \qquad 0 \leq a \leq 1,$$

is convex and nondecreasing.

Proof. It vanishes on $[0, p/2]$. On $[p/2, 1]$ it is $2a^{s+1} - pa^s$, whose first derivative is $a^{s-1}(2(s + 1)a - ps) > 0$ and whose second derivative is $sa^{s-2}(2(s + 1)a - p(s - 1)) \geq 0$. At $p/2$ the right derivative is nonnegative and no smaller than the zero left derivative. This proves both claims on the whole interval. □

4 The mixed-branch theorem

Theorem 4.1 (Mixed rooted triangle branches)**.** *For all integers $r, s \geq 0$ and every graphon W whose edge density lies in $p \in [1/2, 1]$,*

$$\boxed{t(R_{r,s}, W) \geq p^{r+2s+1}(2p - 1) = \Phi_{R_{r,s}}(p).}$$

This is the complete interval required by $\chi(R_{r,s}) = 3$.

Proof. First suppose $s = 0$. By (1), Lemma 3.2, and Jensen,

$$t(R_{r,0}, W) \geq \frac{2p-1}{p} M_{r+2} \geq p^{r+1}(2p-1).$$

Now suppose $s \geq 1$. Under the probability measure ν_r from Lemma 2.1, (1) and Lemma 3.1 give

$$t(R_{r,s}, W) \geq M_r \, \mathbb{E}_{\nu_r} f_{p,s}(A).$$

By Lemmas 3.3 and 2.1,

$$\mathbb{E}_{\nu_r} f_{p,s}(A) \geq f_{p,s}(\mathbb{E}_{\nu_r} A) \geq f_{p,s}(p^2).$$

Since $p \geq 1/2$, one has $p^2 \geq p/2$, and hence

$$f_{p,s}(p^2) = p^{2s}(2p^2 - p) = p^{2s+1}(2p-1).$$

Finally $M_r \geq p^r$ by Jensen (including the identity $M_0 = 1$), so

$$t(R_{r,s}, W) \geq p^{r+2s+1}(2p-1),$$

which equals (2). □

At $p = 1/2$ the target vanishes and every step remains valid. At $p = 1$, $0 \leq W \leq 1$ and $\int W = 1$ imply $W = 1$ almost everywhere, so the density and target both equal 1. No division by $2p - 1$ occurs.

5 Catalogue consequence

Corollary 5.1 (Atlas 95)**.** *NetworkX Atlas graph* 95*, with graph6 code* `EG}?`*, is positive. It has order and size six, chromatic number three, and*

$$\chi_{F_{95}}(x) = x(x-1)^4(x-2).$$

For every graphon W of edge density $p \in [1/2, 1]$,

$$t(F_{95}, W) \geq p^4(2p-1) = \Phi_{F_{95}}(p).$$

Proof. Atlas 95 has NetworkX edge list

$$04,\ 05,\ 12,\ 14,\ 24,\ 34.$$

The vertices $1, 2, 4$ form a triangle. Root it at 4: vertex 3 is a direct leaf, while $4 - 0 - 5$ is a two-edge path, and there are no other edges. Thus the graph is $R_{1,1}$, and Theorem 4.1 applies. The chromatic polynomial and target follow from the formulas in Section 1. □

6 Formalization boundary and exact limitations

The proof uses ordinary, non-injective graphon homomorphism densities on an arbitrary probability space. All integrands are bounded and measurable, so Tonelli–Fubini applies. The only analytic inputs are Jensen, Cauchy–Schwarz, and the displayed opposite-order symmetrization for U. No finite-step reduction, regularity assumption, minimum-degree hypothesis, or injective-copy interpretation is present.

For a Lean formalization, the new reusable statement is Lemma 2.1. Its clean signature should include $p \geq 1/2$ explicitly. The remaining ingredients already occur in the rooted triangle–tree development: the density factorization, pointwise rooted-triangle bound, tail convexity, weighted rooted-triangle inequality, and moment Jensen bound. The graph-specific layer need only exhibit Atlas $95 \cong R_{1,1}$ and transport the generic result.

The theorem covers only a bouquet at one triangle root whose branches have length one or two and are internally disjoint. It does not cover a three-edge path, two length-two paths sharing their intermediate vertex, branches at different triangle vertices, or attachments to a larger clique or triangle book. In particular it does not classify Atlas 97, 100, or 102. The false stronger assertion $t(F + \text{leaf}, W) \geq p\, t(F, W)$ is explicitly refuted in `FAILED_PROOF_ATTEMPTS.md`; it is not used here.

7 What the Lean formalization actually proves

The machine-checked development is `lean/Taeyoung/Methods/MixedBranch.lean`, ending at `satisfiesLowerBour` It proves the single case $(r, s) = (1, 1)$ — the only one of six vertices — and it does so by an argument that is *not* the one above.

The departure: no symmetrization, no Jensen. This note handles general (r, s) by a complement-kernel symmetrization, showing that the mean two-edge-tail density under the d^r-biased measure is at least p^2, and then disposes of all s tails with one convex Jensen step. At $(1, 1)$ neither is needed. Peeling gives

$$t(R_{1,1}, W) = \int \tau\, A\, d, \qquad A = T_W d,$$

and the pointwise Goodman bound $\tau \geq 2A - p$, weighted by $Ad \geq 0$, already reduces the problem to two moments of the single weight $d \cdot A$:

$$t \geq 2C - pB, \qquad B = \int dA, \quad C = \int dA^2.$$

The weighted Cauchy–Schwarz $B^2 \leq pC$ and path Sidorenko $B \geq p^3$ then close it through

$$p\,(2C - pB) - p\,(2p^5 - p^4) \;\geq\; (B - p^3)\big(2(B + p^3) - p^2\big) \;\geq\; 0, \tag{5}$$

both factors being nonnegative because $B \geq p^3 > 0$ and $2p^3 \geq p^2$ on $p \geq 1/2$. Every input predates the row: `Link.rootedTriangle_ge`, `PureChordal.integral_mul_sq_le_integral_mul_integral_mul_sq` with weight d, and `PathSidorenko.pow_three_le_pathIntegral`.

So the two proofs agree on $(1, 1)$ and diverge in generality: this note's is the one that scales in r and s, while (5) is specific to one tail and one leaf. If a seven-vertex member ever enters scope, the symmetrization is the thing to formalize.

Everything else transcribes directly. The peeling is 'Methods/Peeling.lean''s generic ladder plus three collapses; τ appears as its own definition, so the triangle needs no separate lemma.